

# **CASA0023 Remotely Sensing Cities and Environments Learning Diary**

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# Table of contents

|                                              |          |
|----------------------------------------------|----------|
| <b>Preface</b>                               | <b>3</b> |
| About me, . . . . .                          | 3        |
| In this Learning Diary, . . . . .            | 3        |
| <b>1 Introduction</b>                        | <b>4</b> |
| 1.1 Summary . . . . .                        | 4        |
| 1.1.1 Types of Sensor . . . . .              | 4        |
| 1.1.2 Spatial Resolution . . . . .           | 4        |
| 1.1.3 Study Area: Lisboa, Portugal . . . . . | 4        |
| 1.2 Application . . . . .                    | 7        |
| 1.3 Reflection . . . . .                     | 7        |
| 1.4 References . . . . .                     | 7        |
| <b>2 Summary</b>                             | <b>8</b> |
| <b>References</b>                            | <b>9</b> |

# Preface

Hello and welcome to Christy's page on Remote Sensing. This page is an online learning diary for CASA Remotely Sensing Cities and Environments 25/26. This website contains a record of her week-by-week learnings, including summary of key concepts, practical applications, reviews on academic literature and relevant web sources, as well as reflections and discussions on learnt concepts and techniques.

## About me,

She is a current master's student at the Bartlett Centre for Advanced Spatial Analysis (CASA), University College London (UCL) reading MSc, Urban Spatial Science. She holds a bachelor's degree in BSc, Planning and Real Estate from the University of Manchester (2022-2025).

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## In this Learning Diary,

Each chapter will be covering contents taught in the corresponding week:

1. Introduction
  - Getting started with remote sensing, introductions, sensors, spatial resolutions and practical pre-processing
2. Sensor

# 1 Introduction

Remote sensing is not only satellite imagery but a system that allows us to obtain EO data through acquiring information from a distance using sensors on platforms. This section introduces types of sensors and resolutions of remote sensing data. The following study area of Lisboa, Portugal will demonstrate a workflow of analysing Earth Observations (EO) data processed through QGIS and SNAP.

## 1.1 Summary

### 1.1.1 Types of Sensor

**Passive Sensors** detect energy emitted or reflected from the object by the sun. They mainly operate in the visible, infrared, thermal infrared, and microwave bands of the electromagnetic spectrum, mostly used to measure surface temperature, vegetation properties and other physical attributes. However, passive sensors cannot penetrate dense cloud cover, limiting areas with higher cloud rates.

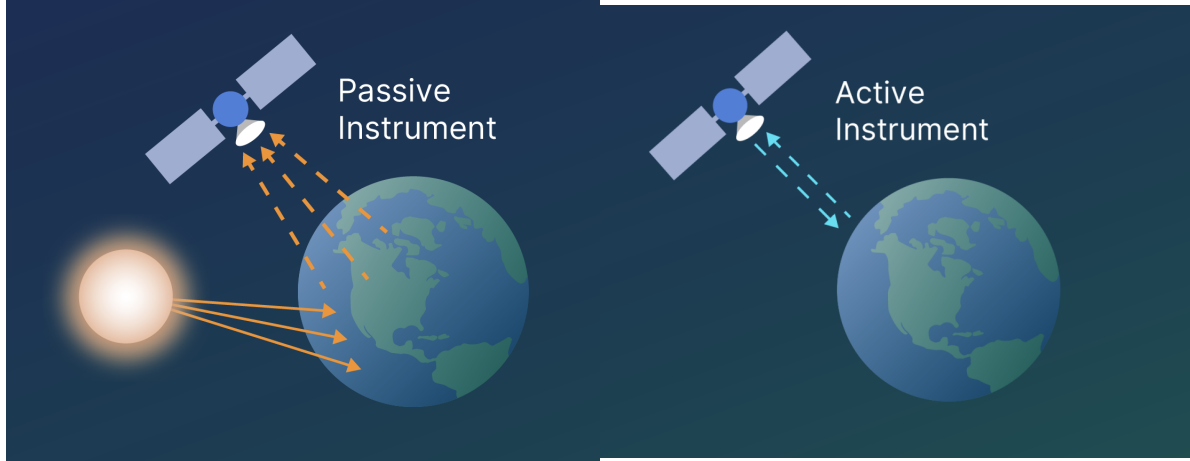
**Active Sensors** emits energy and collect data based on changes in the return signal. Most active sensors have higher capabilities to penetrate atmospheric conditions. For example, the Synthetic Aperture Radar (SAR), has the ability to “see through cloud” as it operates with comparably longer wavelengths, allowing data collections during night hours. This allows them to be less dependent on sunlight which better handles atmospheric constraints.

Overall, observations are shaped by the electromagnetic spectrum and its interactions between radiation, atmosphere, and Earth’s surface, where raditaion can be absorbed, transmitted, or scattered (Tempfli *et al.*, 2009).

### 1.1.2 Spatial Resolution

### 1.1.3 Study Area: Lisboa, Portugal

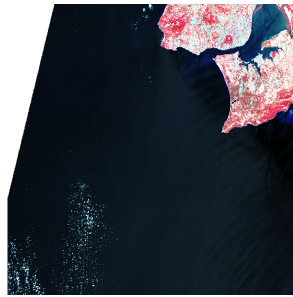
Pre-processing in SNAP



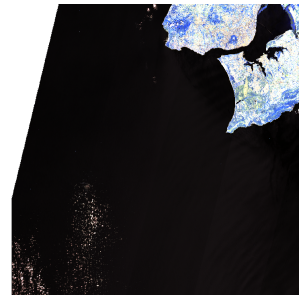
(a) Passive Sensor (Source: NASA EARTHDATA) (a) Active Sensor (Source: NASA EARTHDATA)



(a) True Natural Colour, Sentinel



(a) False Colour Composite, Sentinel



(a) Atmospheric Penetration, Sentinel

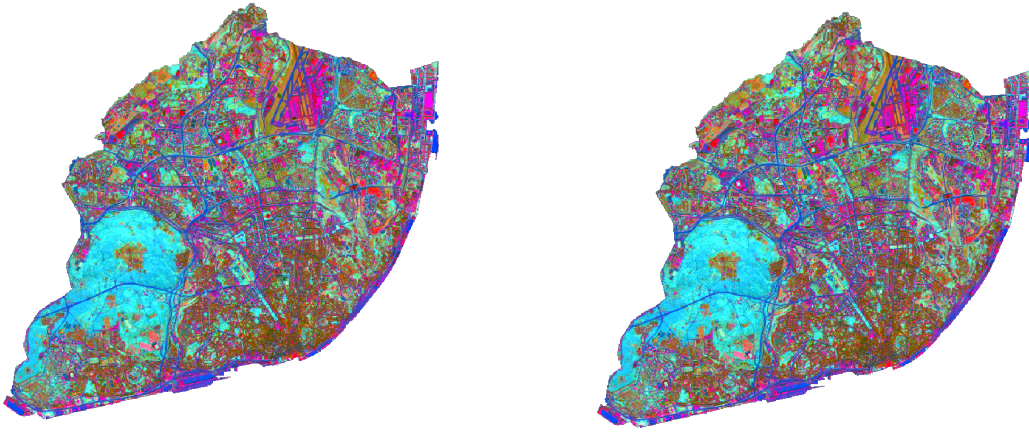
### 1.1.3.1 Tasseled Caps

Maths Band Equations:

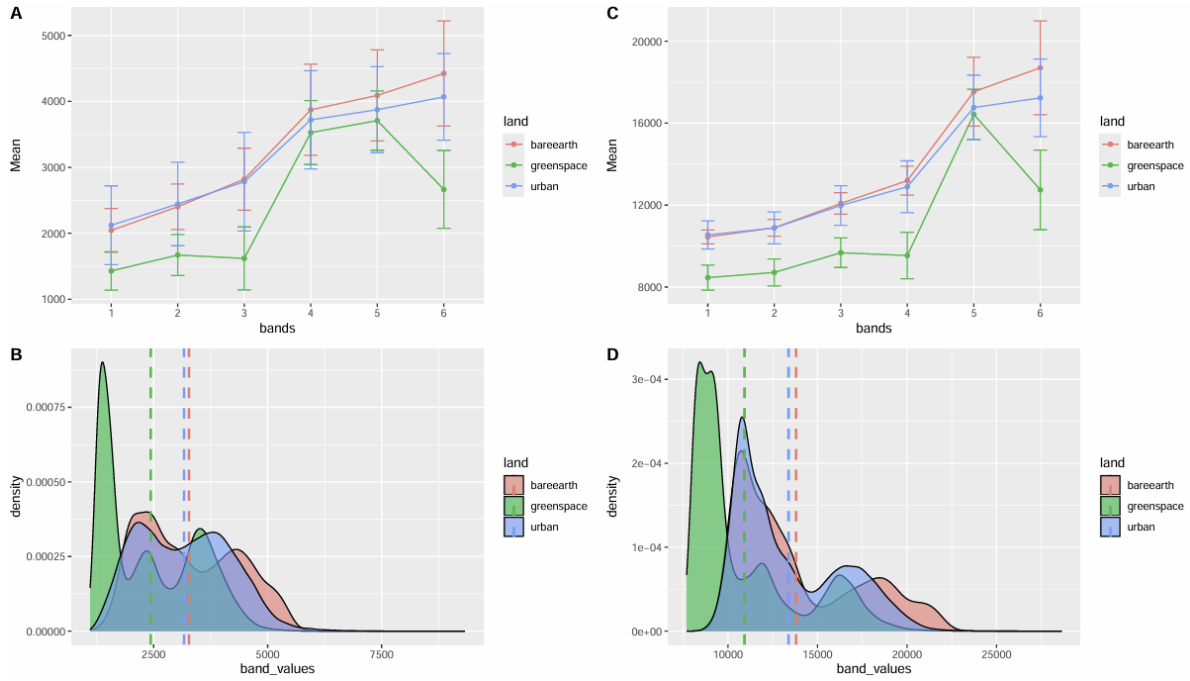
$$\begin{aligned} \textit{Brightness} = & 0.3037(B2) + 0.2793(B3) + \\ & 0.4743(B4) + 0.5585(B8) + \\ & 0.5082(B11) + 0.1863(B12) \end{aligned}$$

$$\begin{aligned} \textit{Greenness} = & -0.2848(B2) - 0.2435(B3) \\ & -0.5436(B4) + 0.7243(B8) + \\ & 0.0840(B11) - 0.1800(B12) \end{aligned}$$

$$\begin{aligned} \textit{Wetness} = & 0.1509(B2) + 0.1973(B3) \\ & +0.3279(B4) + 0.3406(B8) \\ & -0.7112(B11) - 0.4572(B12) \end{aligned}$$



(a) Lisboa in August 2025 (Sentinel, Tasseled Cap Transformed) (a) Lisboa in December 2025 (Sentinel, Tasseled Cap Transformed)



(a) Spectral Reflectance of Sentinel (A, B) and Landsat (C, D)

## 1.2 Application

## 1.3 Reflection

## 1.4 References

Sentinel Data Source:

[https://identity.dataspace.copernicus.eu/auth/realms/CDSE/protocol/openid-connect/auth?client\\_id=sh-5f8b630b-b083-49ed-b340-b8f01ecb81c4&redirect\\_uri=https%3A%2F%2Fbrowser.dataspace.copernicus.eu%2F%3Fzoom%3D5%26lat%3D50.16282%26lng%3D20.78613%26demSource3D%3D%2522MAPZEN%2522%26cloudCoverage%3D30%26dateMode%3DSINGLE&state=26c991f3-8c5e-487e-b23d-d5fe36fbabb7&response\\_mode=fragment&response\\_type=code&scope=openid&nonce=c67a4727-3008-4473-8b84-0a9304c87ce3&code\\_challenge=ZGmOypxnXCivN18EjxS722IF4SDVSsksdcUb\\_zktYp4&code\\_challenge\\_method=S256](https://identity.dataspace.copernicus.eu/auth/realms/CDSE/protocol/openid-connect/auth?client_id=sh-5f8b630b-b083-49ed-b340-b8f01ecb81c4&redirect_uri=https%3A%2F%2Fbrowser.dataspace.copernicus.eu%2F%3Fzoom%3D5%26lat%3D50.16282%26lng%3D20.78613%26demSource3D%3D%2522MAPZEN%2522%26cloudCoverage%3D30%26dateMode%3DSINGLE&state=26c991f3-8c5e-487e-b23d-d5fe36fbabb7&response_mode=fragment&response_type=code&scope=openid&nonce=c67a4727-3008-4473-8b84-0a9304c87ce3&code_challenge=ZGmOypxnXCivN18EjxS722IF4SDVSsksdcUb_zktYp4&code_challenge_method=S256)

## 2 Summary

In summary, this book has no content whatsoever.

1 + 1

[1] 2



## References